

PSYKI – Core

Architecture

Status: canonical. Supersedes `psyai-11c/PSYKI` (v0 sketch) and `mypsai/AgentAgent2` (quarry). Where any repo disagrees with this document, this document wins and the repo gets reshaped.

Provenance: derived from design session 2026-08-21. Nothing here is inherited unexamined from prior authorship.

0. Shape

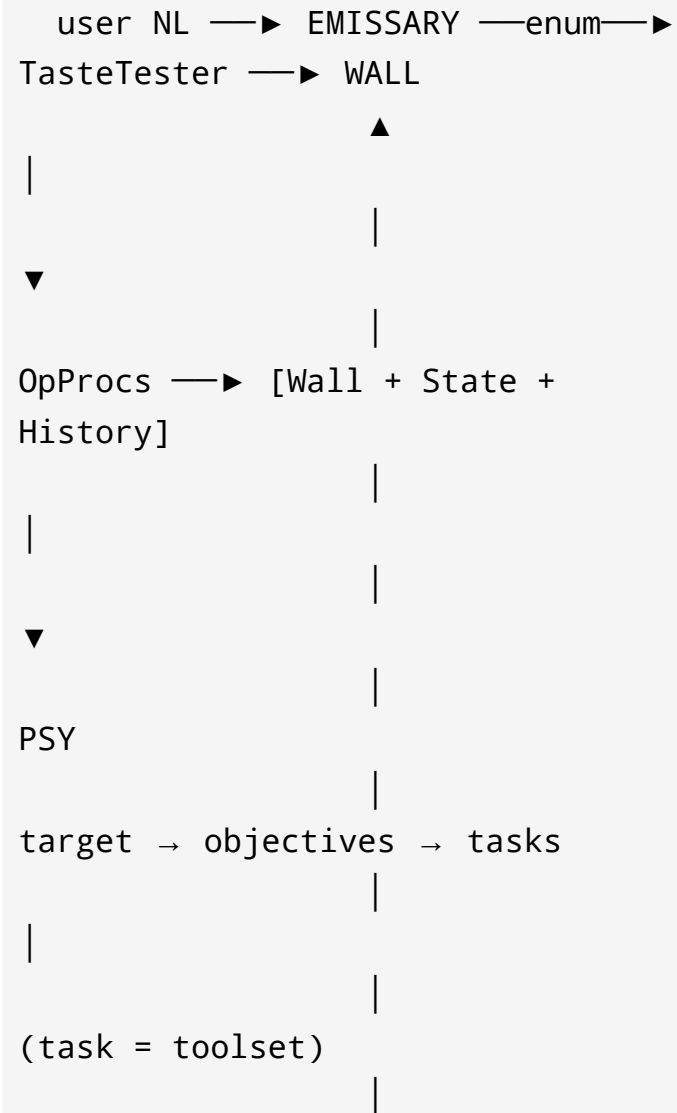
Three reasoning components. Everything else is deterministic code.

Component	Kind	Role
Emissary	model	sole membrane. NL ↔ enum, both directions.
PSY	model	temporal planner. Wall + present + history → target → objectives → tasks.
AgentAgent	model	agent design, code authoring, test authoring. TaskMaster is its subagent.
KI	code	certifying authority. Admits, certifies, revokes. Never reads text.
TasteTester	code	schema + security validation at the Wall write path.

Component	Kind	Role
State	code	bounded projection of all server data. Server core is sole writer.
Wall	code	encrypted, user-authored, append-only.
Log	code	server events + contract records. Bounded depth.
Retinue	code	agent code indexed by toolset signature. Hash-pinned.
OpProcs	code	read-only charter. Hash-verified on every read.

Trust hierarchy: OpProcs > Wall > Log. A directive contradicting OpProcs is voided, not negotiated. A log record never overrides a Wall directive.

1. The cycle





KI — certificate

present state only

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INITIATION



AGENTAGENT

design · code · tests

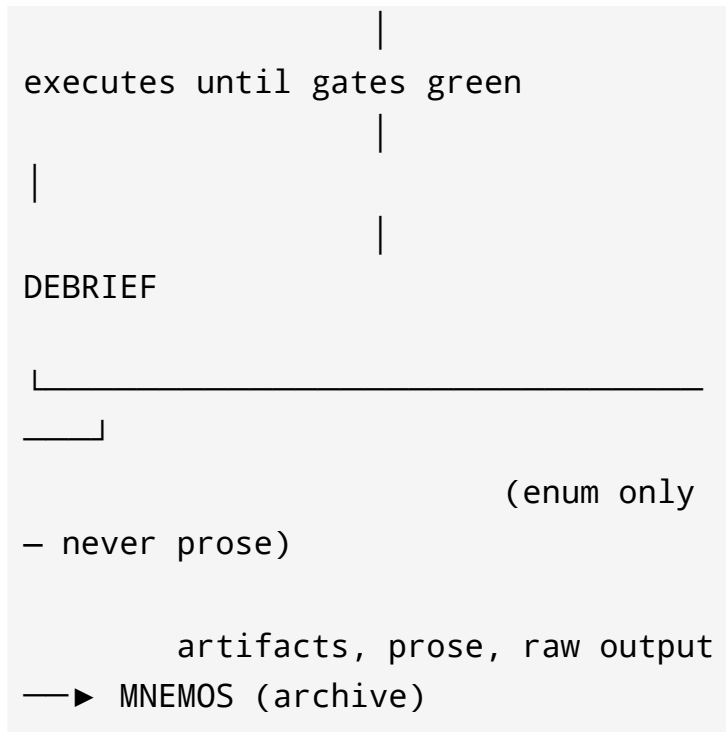
TaskMaster: model binding

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CONTRACT



provisioned agent



The Emissary is the only membrane. Every piece of untrusted text – user input, agent reports, fetched content, mnemos retrieval – crosses here and converts to enum. Nothing else in the system reads free text.

Consequence: the enum protocol is the security boundary, not the model. A

compromised Emissary can only express what the vocabulary allows.

2. Invariants

#	Invariant
I1	PSY's context is exactly (Wall, State, History). Sealed; audited byte-wise.
I2	KI's context is exactly (State, Plan, Event). No history, no future.
I3	KI is a pure total function. No RNG, no wall-clock, no model call.
I4	No prose crosses any internal boundary. Enum + typed fields only.
I5	OpProcs is read-only and hash-pinned. The server cannot author its charter.
I6	Only TasteTester-admitted bytes reach the Wall.

#	Invariant
I7	Every objective carries a traceable <code>directive_id</code> . Orphans rejected at plan validation.
I8	A task is bounded by its toolset. Fan-out is internal to a task.
I9	Certificates are issued and revoked by KI alone.
I10	Both PSY and KI operate under a dynamic context choke $\leq$ server capacity.
I11	The server core is the sole writer of State. Components emit events; they never mutate State.
I12	KI reads a frozen State snapshot at tick. Determinism is over the snapshot, not the live struct.

3. Task boundary rule

A task is defined by its toolset. Same tools
= one task, regardless of
target count. Testing 1 model and testing
200 models are the same task with
different fan-out. The moment the required
toolset changes, it is a new task.

This makes task decomposition
deterministic rather than a judgment call,
and
makes agent provisioning nearly free: the
agent *is* the toolset plus a purpose.

The tool manifest therefore partitions the
task-type space.

4. Enum protocol

Three vocabularies. Enums carry the **frame**;
typed fields carry the
content. String fields exist but are never
interpolated as instruction.

4.1 Intent (Emissary → Wall)

```
VERB      := CREATE | MODIFY |  
ANALYZE | TEST | RESEARCH  
          | DEPLOY | REPAIR |  
DOCUMENT | REMOVE | EVALUATE  
SCOPE     := FILE | MODULE | REPO |  
SERVICE | DATASET | MODEL | SYSTEM  
URGENCY   := DEFER | NORMAL |  
PRIORITY  | IMMEDIATE
```

Emissary hard rule: if intent cannot be encoded, it *refuses and re-asks*.

It never approximates. A silently lossy

Emissary is the worst failure this design admits, because nothing downstream can detect it.

4.2 Contract (AgentAgent → provisioned agent)

The contract is simultaneously the agent's lifecycle, sole purpose, and prompt.

```
Contract {  
  contract_id  
  certificate_id          # KI-
```

```
issued; invalidated on fulfillment
  directive_id          # lineage
to Wall – I7
  objective_id

  toolset_signature     # defines
the task – I8
  fanout_targets[]      # 1..N;
internal to the task

  agent_ref             # retinue
hash-pin, or NEW
  model_binding         #
TaskMaster-assigned
  capability_floor      # context,
modality, tool-calling

  completion_predicate  # see
below
  gate_chain            #
format→lint→typecheck→test→covera
ge→security
  eval_threshold        # default
0.90; safety = 1.0 always
```

```

    retry_budget
    return_address          # Emissary
    debrief channel
}

COMPLETION := ALL | THRESHOLD(n) |
BEST_EFFORT | FIRST_SUCCESS

```

Completion predicate is set at authoring time, not decided at debrief. This is what resolves partial failure inside a fan-out task.

4.3 Verdict (agent → Emissary → Log)

```

VERDICT := FULFILLED | PARTIAL |
FAILED_GATE | FAILED_TOOL
          | FAILED_BUDGET | STALLED
| REJECTED | ESCALATE

VerdictRecord {
    contract_id
    verdict
    gates_passed[]  gates_failed[]
}

```

```
eval_score
fanout_completed / fanout_total
anomaly_flags[]
artifact_refs[]          # pointers
into mnemos – never inline
}
```

For code-shaped contracts the debrief **is** the test report. Already enum-shaped, needs no interpretation.

5. Lifecycle

1. KI issues certificate → task becomes Initiation.
2. AgentAgent designs; TaskMaster binds model → Contract.
3. Agent executes until gate chain green or retry budget exhausted.
4. Debrief crosses the Emissary as enum → Log.
5. **KI revokes the certificate.** Model returns to pool. Agent code returns to

Retinue, hash-pinned and versioned.

Prose and artifacts go to mnemos.

The contract *is* the agent's lifespan.

Nothing survives it but the record.

6. Escalation tiers

Tier	Trigger	Handler
1	contract fails gates	AgentAgent retries within retry budget
2	tier-1 budget exhausted	PSY replans the objective
3	replan counter exceeded	Emissary returns to the user

Tier 3 is the termination condition for the closed loop. The cycle

Emissary → Wall → PSY → KI → AgentAgent → exec
has no human in it; the
replan counter is what guarantees it halts.

Ambiguous debrief also escalates to tier 3.

There is no user to interrogate on the return path, so the Emissary surfaces rather than guesses.

6a. State

State is derived from **everywhere** and materialized as **one bounded struct**.

It is a projection, not a collection.

Components write into it by emitting events to the server core; nobody reads its sources through it.

Every field is scalar or enum. No prose ever enters State.

```
State {  
    state_rev                #  
    monotonic; the freshness anchor  
    opprocs_hash            # I5  
    verification  
  
    wall_rev                 # revision
```

```

number ONLY – never Wall contents
  log_head
  recent_verdicts[]          # last N,
enum only

  certificates_outstanding[]
  agent_pool_available[]
  model_residency            # which
model is loaded now
  toolset_locks[]           # tools
currently held by a live contract
  contract_queue_depth

  ceilings { gpu, memory, disk,
context }
}

```

Who reads what:

- PSY reads Wall contents directly, plus State, plus History. (I1)
- KI reads State, Plan, Event. Never Wall, never Log, never prose. (I2)
- The Emissary reads State only to report status back to the user.

The freshness check. PSY plans against `state_rev = N`. KI ticks at `state_rev = M`. If any precondition changed between N and M, KI refuses admission or revokes a running certificate. This is the entire justification for KI seeing only the present, and it is why the revision counter exists.

Write discipline. Concurrent mutation on the one structure KI's determinism depends on would be fatal, so the server core owns it: components emit events, the core folds them into State and increments `state_rev`, KI takes a frozen snapshot at tick. Determinism is over the snapshot. (I11, I12)

7. Memory discipline

- **Log** holds contract-level records only: verdict, deltas, counts, anomaly

flags, pointers. Small, enum, bounded to a few sessions.

- **Mnemos** holds everything else: prose, artifacts, raw output. Unbounded.
 - **Rule:** if PSY ever needs to read mnemos to understand the present, the split is wrong.
 - Mnemos retrieval is **explicitly requested only** — PSY cannot trigger it. Retrieved content re-enters through the Emissary as enum, like any ingress.
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8. Model binding

Role → model is configuration, not architecture. Swapping a model does not ripple because the enum protocol isolates it.

Role	Candidate	Rationale
PSY	VibeThinker-3B	deep single-shot reasoning, no dialogue, no tools
Emissary	Gemma 4 E4B	multimodal, conversational; disambiguation is a dialogue
AgentAgent	largest available + escalation hook	heaviest role; design quality gates everything

AgentAgent authors the design and the tests. A smaller model runs the implement loop until green. This is what makes local models viable — it converts "write correct code" into "iterate until the oracle says yes." Keep

the escalation path for contracts that exhaust retries on genuinely hard work.

Residency is a schedule, not an allocation.

The pipeline is sequential;
model swap cost per phase is an unmeasured number that determines whether an objective loop takes seconds or minutes. Measure it early.

9. Port manifest

From AgentAgent2 — take

- `AGENTAGENT2.system.md` — **truncated**: enters at DESIGN, exits at DELIVER. Its INTENT and PLAN phases are satisfied externally by the Contract. Its DEBRIEF becomes a boundary crossing, not a terminal phase.
- `skills/ (9)` — especially `subagent-orchestration` → TaskMaster

- `style/` – 5-language style engine, keep whole
- `tools/tool_manifest.json` – becomes the task-type partition
- `design/schemas/agent_design.schema.jsor`
- `templates/agent_project_scaffold/` – gate-ready skeleton

From AgentAgent2 – leave

Duplicate (1) files, `agentagent2-complete.zip`, session-event JSON at public root, `verify_report*.json`. Do not inherit the hygiene.

From PSYKI v0 – take

- OpProcs concept (read-only hash-pinned charter)
- KI determinism harness and CI gate
- Lateral-inhibition fix in the BG gate (Buridan deadlock)

- Sealed-context whitespace → word-boundary audit fix
- Wall append-only + hash chain — **but replace blake2b-XOR with XChaCha20-Poly1305 before the Wall takes a second writer**

From PSYKI v0 — leave

- The corpus/database layer
 - `NullPlanner` beyond its role as ablation control
 - Regex-only taste test as a boundary (keep as speed bump with telemetry)
 - The v0 module map, which never matched the tree
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10. Open

1. **Model swap cost** — unmeasured. Gates the whole latency picture.

2. **State event ordering** — resolved that the core is sole writer (§6a). Open: whether the event fold is strictly ordered or batched per tick, and what happens to events arriving mid-tick.
3. **Retinue versioning** — a tool changing under a cached agent is a nasty silent failure. Needs a tool-version component in the signature.
4. **Auto-evolution** — deferred. A system with a return path that rewrites its own settings observes its own effects. Gate behind the replan counter and the escalation tiers, not just metrics. OpProcs stays immutable regardless.